Cisco Packet Tracer - VLAN Configuration and Network Segmentation

Portfolio Project

Project Overview

Designed and configured a small enterprise network in Cisco Packet Tracer using Virtual Local Area Networks (VLANs) to segment traffic and improve network organization and security.

Technologies Used

- Cisco Packet Tracer
- Cisco IOS CLI
- VLAN Configuration
- TCP/IP Networking
- Layer 2 Switching

Key Tasks

- Created VLAN 10 (SALES) and VLAN 20 (HR).
- Assigned switch ports to appropriate VLANs.
- Configured end devices with static IPv4 addresses.
- Verified VLAN membership using Cisco IOS commands.
- Tested network communication to confirm VLAN isolation.
- Demonstrated how VLANs separate broadcast domains and enhance network security.

Network Diagram

```
Cisco Packet Tracer VLAN Lab +----------------------+ | Cisco 2960 | | Switch |
+----------------------+ | | Fa0/1| |Fa0/13 | | +-----------+ +-----------+ | PC0 |
| PC1 | | VLAN 10 | | VLAN 20 | | SALES | | HR | +-----------+ +-----------+ IP:
192.168.10.10 IP: 192.168.20.10 VLAN 10 and VLAN 20 are isolated from each other
until inter-VLAN routing is configured.
```

Configuration Commands

```
enable configure terminal vlan 10 name SALES vlan 20 name HR interface fa0/1
switchport mode access switchport access vlan 10 interface fa0/13 switchport mode
access switchport access vlan 20
```

Results

VLAN 10 devices successfully connected within their VLAN. VLAN 20 devices successfully connected within their VLAN. Communication between VLANs was blocked without inter-VLAN routing, confirming proper network segmentation.

Skills Demonstrated

Network Administration, Cisco Switch Configuration, VLAN Implementation, IP Addressing, CLI Troubleshooting, and Network Segmentation.